Jai Sharma

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EDUCATION

University of British Columbia - *Master of Data Science (Computational Linguistics) - 90% Average* June 2026
Relevant Coursework: Supervised Learning II, Computing Platforms, Statistical Inference, Algorithms & Data Structures
University of Western Ontario - *Bachelor of Computer Science (GPA: 3.8/4.00)* June 2025

SKILLS

Programming Languages: Python, R, Java, Golang, C++, C, JavaScript, Scala, C#
Data Science & ML: Pandas, NumPy, Scikit-learn, LLMs, Generative AI, Seaborn, Matplotlib, Random Forest, XGBoost
Tools & Platforms: SQL, Linux, Bash, Git, PySpark, PyTorch, Hadoop, Apache Spark, AWS, Azure, TensorFlow, Docker

PROJECTS

Equity Drawdown Risk Prediction Pipeline | *Financial ML, Time Series, Risk Modeling* March 2026

- Built an end-to-end machine learning pipeline to rank stocks by near term drawdown risk, helping identify which names were most likely to suffer major losses over the next 60 trading days.
- Combined price behaviour, market relative signals, short interest, and fundamentals into a repeatable research workflow covering 1.03M+ observations, with careful time-based testing to better reflect real investment use.
- Delivered a practical risk screening tool that achieved 2.31x lift in the top decile on the held-out test set, showing strong value for prioritizing higher risk names for deeper review.

Predicting Momentum Shifts in Basketball using Supervised Learning | *NLP*, *Feature Engineering* Jan 2026

- Built an end-to-end NLP and machine learning pipeline that transformed raw event logs from NBA games play-by-play data into possession-level features and improved prediction performance from a 0.342 macro F1 baseline to 0.786 macro F1 on held-out games.
- Designed a reproducible data pipeline for scraping, cleaning, labeling, feature engineering, and grouped model evaluation across 415 NBA games and 80K+ possessions.
- Identified score margin and recent possession history as the strongest predictors of possession scoring, outperforming a majority-class baseline with a hybrid text + gradient boosting model.

Spotify Hit Song Prediction: End-to-End ML Pipeline| *Predictive Modeling, Supervised Learning* *Oct* 2025

- Built an end-to-end ML pipeline that enriched 2,000+ Spotify tracks via the Web API and predicted song popularity using a DummyClassifier baseline, Logistic Regression, and an RBF-kernel SVM (87% accuracy, 0.91 ROC-AUC).
- Analysis showed artist popularity was the strongest predictor of hit status; features like valence, release year, energy, loudness, and instrumentalness also mattered, with hits tending to be more energetic, brighter, and more danceable.

EXPERIENCE

L3Harris Technologies May 2023 – Aug 2024
Software Engineer Co-op *Hamilton, Ontario*

- Designed and implemented automated testing using pytest and integrated tests into CI/CD pipelines, identifying and resolving 17 critical defects and reducing bugs by 11% in an agile environment.
- Developed and optimized C/C++ systems across kernel, OS, and application layers for multi-core embedded platforms, improving system reliability and performance.
- Improved DevOps CI/CD pipelines using Jenkins and Git, reducing build times by 15.6% and increasing release reliability through automated build validation.

Bell Canada May 2022 – Aug 2022
Software Developer Intern *Mississauga, Ontario*

- Upgraded PHP, HTML/CSS, and Twig applications for Bell's media sites, reducing front-end bugs by 18% and improving page load times by 15%.
- Improved frontend performance and reduced bugs in media platforms, contributing to better content delivery and user engagement.
- Investigated and resolved environment-specific integration issues to ensure consistent behavior across deployments.